PHARMACOLOGY

Laxatives and Anti-Diarrhoeal Agents

LG 3
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GIT BLOCK

Pharmacology GIT Block · MEDucation by Mattin Majid

BEFORE YOU READ — ANSWER FROM MEMORY

1. Why are laxatives contraindicated in an acute surgical abdomen or organic GIT obstruction?
2. What distinguishes the mechanism of bulk laxatives from osmotic/saline laxatives?
3. Why is bismuth subsalicylate the first-line choice for traveler's diarrhoea, and what harmless side effect does it cause?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Constipation — definition & first-line management

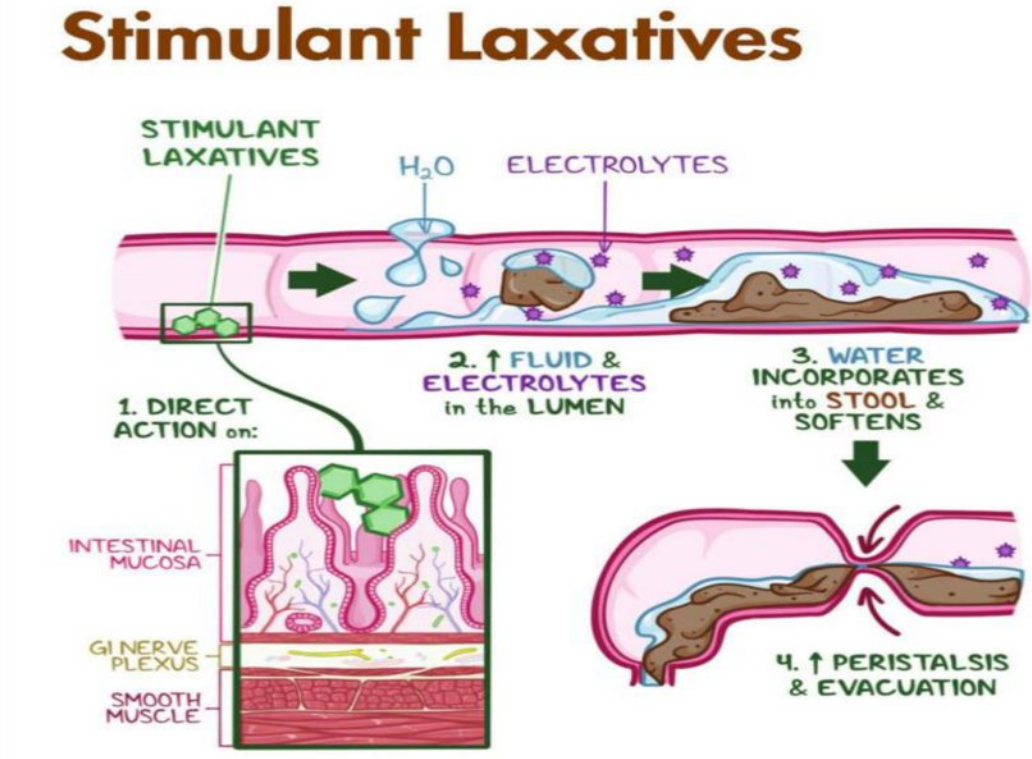
Constipation is a reduced frequency or ease of stool passage relative to what is normal/expected for that individual. First-line non-drug therapy: fibre-rich diet, increased hydration, minimizing tea/coffee, and physical exercise (activates abdominal muscles and peristalsis).

2 Laxatives — core principles, indications & contraindications

Risk / Use	Detail
Drug loss	Accelerated transit → loss of effect for poorly absorbed, delayed-acting, or extended-release oral drugs
Electrolyte balance	Chronic/excessive use → significant electrolyte imbalance
Dependency	Prolonged reliance → high risk of dependency, loss of normal motility
Indications	Short-term constipation; toxic substance excretion; diagnostic prep (X-ray/colonoscopy); anal fissures/ haemorrhoids; post-haemorrhoidectomy
Contraindications	Undiagnosed abdominal pain; IBD (perforation risk); organic GIT obstruction — absolute contraindication

Warning: laxatives in an acute surgical abdomen can cause fatal intestinal rupture.

3 Laxative classes

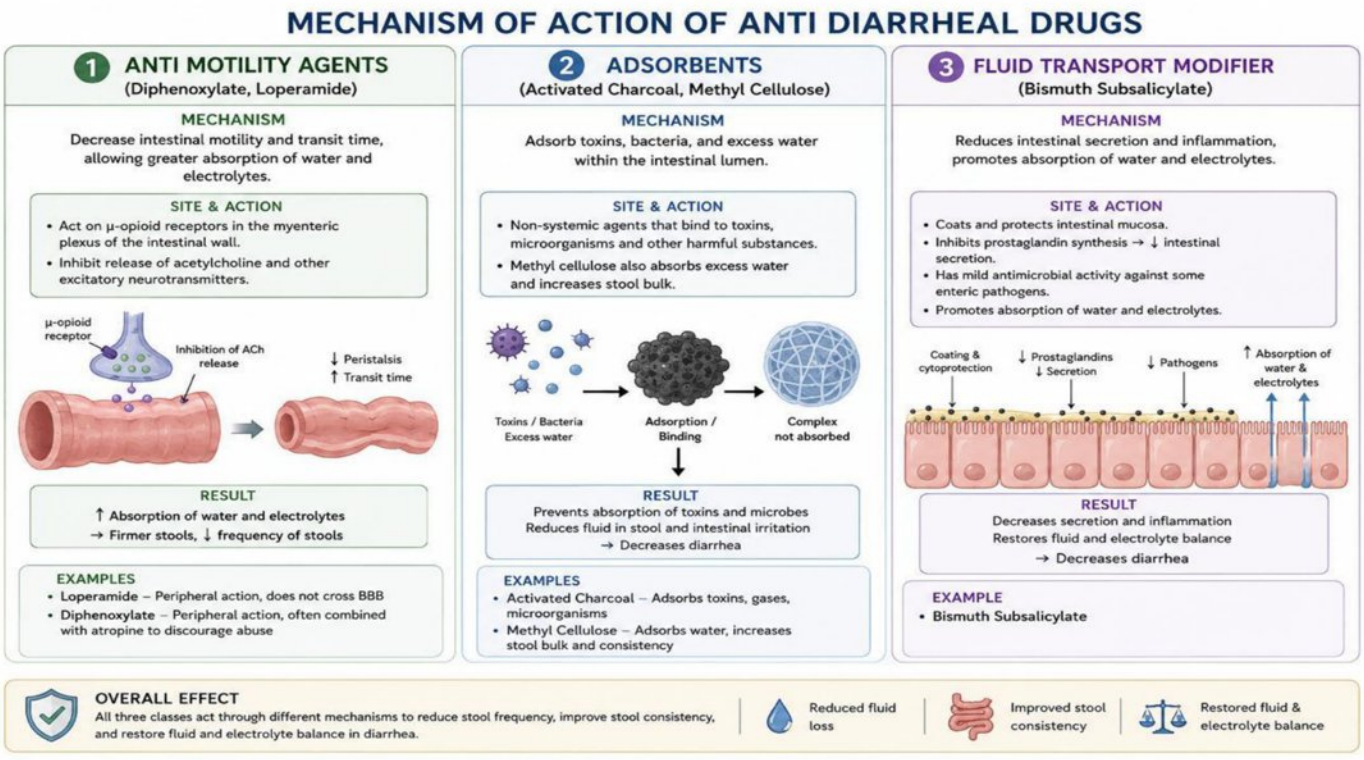


Stimulant laxatives act directly on GI nerve plexus/smooth muscle to increase fluid/electrolyte secretion and peristalsis, driving evacuation.

Irritants/ stimulants (senna, bisacodyl, castor oil)	Senna: bowel evacuation in 8–10h, promotes active water/electrolyte secretion. Bisacodyl: potent colon stimulant acting on nerve fibers in colonic mucosa. Castor oil: hydrolyzed to ricinoleic acid (irritant) — contraindicated in pregnancy (stimulates uterine contractions).
Bulk laxatives (methylcellulose, psyllium)	Hydrophilic colloids form gels in the large intestine → water retention, bowel distension → reflexively stimulates peristalsis. Caution in bedridden/immobile patients — high risk of mechanical obstruction.
Saline & osmotic (Mg citrate/ hydroxide, PEG, lactulose)	Mg salts: nonabsorbable, hold fluid via osmosis → distension/defecation within hours. PEG: electrolyte solution for colonic lavage before radiological/endoscopic procedures. Lactulose: colonic bacteria metabolize it to lactic/formic/acetic acids → ↑osmotic pressure → fluid accumulation, soft stools.
Stool softeners (docusate)	Surfactant, emulsifies with stool → softer faeces. Takes days to work — used for prophylaxis, not acute treatment.
Lubricants (mineral oil, glycerin suppositories)	Lubricate intestinal wall, ease passage of hard stool. Oral mineral oil must be taken upright — risk of aspiration/lipid pneumonia if not.
Chloride channel activators (lubiprostone)	Only agent in class. Activates luminal Cl^- channels → fluid secretion. Minimal systemic electrolyte change; approved for chronic constipation with no tolerance/dependency risk; rapidly metabolized in stomach/jejunum (few interactions).

4 Anti-diarrhoeal therapy — pathophysiology & approach

Major pathophysiologic drivers of diarrhoea: **increased GI motility** and **decreased fluid absorption**. Three core management approaches in severe acute diarrhoea: **(1) fluid balance** — maintain fluid/electrolyte equilibrium (ORS); **(2) anti-infectives** — specific antimicrobials for infectious cause; **(3) spasmolytics** — motility reduction and symptomatic antidiarrhoeal agents.



Three antidiarrhoeal drug classes and their mechanisms — antimotility agents, adsorbents, and the fluid transport modifier bismuth subsalicylate.

5 Anti-diarrhoeal drug classes

Antimotility agents (diphenoxylate, loperamide)	Opioid analogs (meperidine-related). Stimulate presynaptic opioid receptors in enteric nervous system → inhibit ACh release → ↓peristalsis. No systemic analgesia at clinical doses. Loperamide: peripheral action, doesn't cross BBB. Diphenoxylate: combined with atropine (Lomotil) to discourage abuse. **Contraindicated in young children and severe colitis — toxic megacolon risk.**

Adsorbents (activated charcoal, methylcellulose, aluminium hydroxide)	Bind toxins/microorganisms/excess water in the lumen (non-systemic). Methylcellulose also increases stool bulk.
Fluid transport modifier (bismuth subsalicylate)	Adsorbs toxins/microorganisms, coats/protects mucosa, decreases fluid secretion (salicylate anti-inflammatory action). **First-line for traveler's diarrhoea.** Significantly less effective than antimotility agents; can interfere with absorption of co-administered oral drugs. Harmless darkening of tongue/stool (bismuth sulfide).

P Must-remember pearls

- Laxatives are absolutely contraindicated in organic GIT obstruction and undiagnosed abdominal pain — use in an acute surgical abdomen can cause fatal intestinal rupture.
- Castor oil is contraindicated in pregnancy (stimulates uterine contractions); bulk laxatives risk mechanical obstruction in bedridden/immobile patients.
- Oral mineral oil must be taken upright to avoid aspiration and lipid pneumonia.
- Docusate (stool softener) takes days to work — it's for prophylaxis, not acute relief.
- Lubiprostone is the only chloride-channel-activator laxative; unlike most classes, it carries no tolerance/dependency risk and is rapidly metabolized in the stomach/jejunum (minimal interactions).
- Loperamide and diphenoxylate are opioid analogs that inhibit ACh release via presynaptic opioid receptors — contraindicated in young children and severe colitis (toxic megacolon risk). Loperamide doesn't cross the BBB, so no systemic analgesia.
- Diphenoxylate is combined with atropine (Lomotil) specifically to discourage abuse.
- Bismuth subsalicylate is first-line for traveler's diarrhoea but less effective than antimotility agents, and can interfere with absorption of co-administered drugs; it harmlessly darkens the tongue/stool via bismuth sulfide.

Q Test yourself

1. A pregnant patient asks about using castor oil for constipation. Why is this contraindicated in pregnancy?

A. It stimulates uterine contractions
B. It causes severe electrolyte depletion within hours
C. It is teratogenic in the first trimester
D. It crosses the placenta and sedates the fetus
E. It causes maternal hepatotoxicity

2. A bedridden, immobile patient is prescribed a bulk laxative such as psyllium. What specific risk should be considered in this patient?

A. Risk of toxic megacolon
B. High potential for mechanical intestinal obstruction
C. Risk of severe uterine contractions
D. Risk of lipid pneumonia from aspiration
E. Risk of permanent loss of intestinal motility

3. A patient is instructed to take oral mineral oil in an upright position. What is the reason for this instruction?

A. To prevent reflux and oesophagitis
B. To improve its laxative efficacy
C. To prevent aspiration into the lungs and severe lipid pneumonia
D. To prevent electrolyte imbalance
E. To reduce the risk of toxic megacolon

4. A patient with severe ulcerative colitis is prescribed loperamide for diarrhoea. Why is this contraindicated in this setting?

A. High risk of severe hepatotoxicity
B. High risk of opioid dependence at antidiarrheal doses
C. High risk of causing uterine contractions
D. High risk of triggering toxic megacolon
E. High risk of causing lipid pneumonia

5. A traveler develops diarrhoea abroad and wants a first-line over-the-counter treatment. Which agent does the lecture specifically name as the first-line choice for traveler's diarrhoea?

A. Activated charcoal **B.** Diphenoxylate + atropine **C.** Methylcellulose **D.** Lactulose **E.** Bismuth subsalicylate

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1 — A — Castor oil is hydrolyzed to ricinoleic acid, an irritant that increases peristalsis — it is contraindicated in pregnancy because it stimulates uterine contractions.

2 — B — Bulk laxatives form gels that cause water retention and bowel distension; in bedridden/immobile patients, this carries a high potential for mechanical intestinal obstruction.

3 — C — Oral mineral oil must be taken upright to prevent aspiration into the lungs, which can cause severe lipid pneumonia.

4 — D — Antimotility agents (diphenoxylate, loperamide) are contraindicated in young children and patients with severe colitis due to the high risk of triggering toxic megacolon.

5 — E — Bismuth subsalicylate (Pepto-Bismol) is named as the first-line choice for traveler's diarrhoea, working via salicylate anti-inflammatory action and mucosal coating.

SPACED REVIEW

☐ Day 1	☐ Day 3	☐ Day 7	☐ Day 21	☐ Pre-exam

Pharmacology GIT Block · MEDucation by Mattin Majid · Source: Dr. Nermin Jamal lecture slides, LG 3 (full deck); references Lippincott Illustrated Reviews: Pharmacology 7th Ed and Rang & Dale's Pharmacology 8th Ed.
Study alongside the lecture, not instead of it.